

Worksheet 17

November 23, 2026

Name: _____

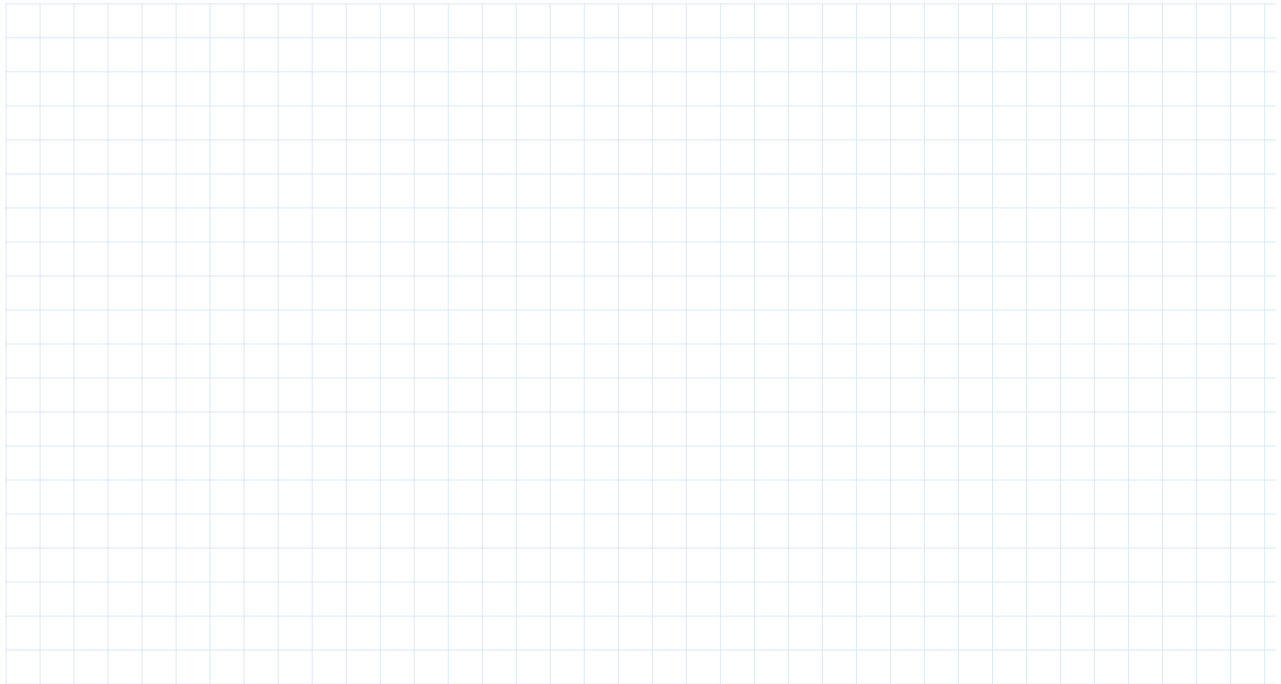
About 10 minutes total. Begin independently, then discuss. Show your work.

1. Rewrite in sine and cosine

Prove the identity wherever both sides are defined:

$$\frac{\sec x - \cos x}{\tan x} = \sin x.$$

(a) Start from the left side and show a sequence of equivalent expressions.



Worksheet 17

November 23, 2026

Name: _____

About 10 minutes total. Begin independently, then discuss. Show your work.

2. Use a conjugate

Prove the identity wherever both sides are defined:

$$\frac{\sin x}{1 + \cos x} = \frac{1 - \cos x}{\sin x}.$$

(a) Start from the left side and multiply by a useful form of 1.

